

Ex. No. : 02

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Calculator

Aim

Develop a scientific calculator to perform arithmetic and mathematical functions using `Math` class. [Your scientific calculator should contain +, *, /, =, cos, sin, tan, pow, sqrt, log, tan and mod].

Procedure:

Step 1 : File -> NewProject

Provide the application name and Click "Next"

Step 2 : Select the target android devices

Select the minimum SDK to run the application. Click "Next".

Step 3 : Choose the activity for the application (By default choose "Blank Activity").

Click "Next".

Step 4 : Enter activity name and click "Finish".

Step 5 : Edit the program.

Step 6 : Run the application, 2-ways to run the application.

1. Running through emulator

2. Running through mobile device

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Calculator"
        tools:targetApi="31">
        <activity android:name=".MainActivity"
            android:exported="true"
            android:windowSoftInputMode="stateHidden">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/display"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="32sp"
        android:gravity="end"
        android:inputType="none"
        android:focusable="false"
        android:cursorVisible="false"
        android:text="0"/>

    <GridLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:columnCount="4"
        android:rowCount="5"
        android:columnWidth="80dp"
        android:useDefaultMargins="true">

        <!-- Row 1 -->
        <Button
            android:id="@+id/btnClear"
            android:text="C"
            android:layout_columnWeight="1"
            android:layout_rowWeight="1"
            style="@style/CalculatorButton"/>

        <Button
            android:id="@+id/btnPlusMinus"
            android:text="+/-"
            android:layout_columnWeight="1"
            android:layout_rowWeight="1"
            style="@style/CalculatorButton"/>
```

```

<Button
    android:id="@+id/btnPercent"
    android:text="%"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btnDivide"
    android:text="/"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorOperatorButton"/>

<!-- Row 2 -->
<Button
    android:id="@+id/btn7"
    android:text="7"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btn8"
    android:text="8"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btn9"
    android:text="9"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btnMultiply"
    android:text="×"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorOperatorButton"/>

<!-- Row 3 -->

```

```

<Button
    android:id="@+id/btn4"
    android:text="4"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btn5"
    android:text="5"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btn6"
    android:text="6"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btnSubtract"
    android:text="-"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorOperatorButton"/>

<!-- Row 4 -->
<Button
    android:id="@+id/btn1"
    android:text="1"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btn2"
    android:text="2"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btn3"

```

```

        android:text="3"
        android:layout_columnWeight="1"
        android:layout_rowWeight="1"
        style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btnAdd"
    android:text="+"
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorOperatorButton"/>

<!-- Row 5 -->
<Button
    android:id="@+id/btn0"
    android:text="0"
    android:layout_columnWeight="2"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btnDecimal"
    android:text="."
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorButton"/>

<Button
    android:id="@+id/btnEquals"
    android:text="="
    android:layout_columnWeight="1"
    android:layout_rowWeight="1"
    style="@style/CalculatorOperatorButton"/>
</GridLayout>
</LinearLayout>

```

MainActivity.kt

```
package com.example.calculator

import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat

class MainActivity : AppCompatActivity() {

    private lateinit var display: EditText
    private var currentInput = StringBuilder()
    private var currentOperator: String? = null
    private var firstOperand: Double? = null

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main) // Changed to activity_main

        display = findViewById(R.id.display)

        // Number buttons
        val numberButtons = listOf(
            R.id.btn0, R.id.btn1, R.id.btn2, R.id.btn3, R.id.btn4,
            R.id.btn5, R.id.btn6, R.id.btn7, R.id.btn8, R.id.btn9
        )

        numberButtons.forEach { buttonId ->
            findViewById<Button>(buttonId).setOnClickListener {
                appendNumber((it as Button).text.toString())
            }
        }

        // Operator buttons
        val operatorButtons = listOf(
            R.id.btnAdd, R.id.btnSubtract, R.id.btnMultiply, R.id.btnDivide
        )

        operatorButtons.forEach { buttonId ->
            findViewById<Button>(buttonId).setOnClickListener {
                setOperator((it as Button).text.toString())
            }
        }
    }
}
```

```

    }
}

// Other buttons
findViewById<Button>(R.id.btnDecimal).setOnClickListener { appendDecimal() }
findViewById<Button>(R.id.btnClear).setOnClickListener { clearAll() }
findViewById<Button>(R.id.btnEquals).setOnClickListener { calculateResult() }
findViewById<Button>(R.id.btnPlusMinus).setOnClickListener { toggleSign() }
findViewById<Button>(R.id.btnPercent).setOnClickListener { applyPercent() }
}

private fun appendNumber(number: String) {
    if (currentInput.isEmpty() && number == "0") return // Prevent leading zeros

    currentInput.append(number)
    updateDisplay()
}

private fun appendDecimal() {
    if (currentInput.isEmpty()) {
        currentInput.append("0.")
    } else if (!currentInput.contains(".")) {
        currentInput.append(".")
    }
    updateDisplay()
}

private fun clearAll() {
    currentInput.clear()
    firstOperand = null
    currentOperator = null
    updateDisplay()
}

private fun setOperator(operator: String) {
    if (currentInput.isNotEmpty()) {
        firstOperand = currentInput.toString().toDouble()
        currentOperator = operator
        currentInput.clear()
    } else if (firstOperand != null) {
        // Allow changing the operator without new input
        currentOperator = operator
    }
}

```



```

}

private fun calculateResult() {
    if (firstOperand != null && currentOperator != null && currentInput.isNotEmpty()) {
        val secondOperand = currentInput.toString().toDouble()
        val result = when (currentOperator) {
            "+" -> firstOperand!! + secondOperand
            "-" -> firstOperand!! - secondOperand
            "x" -> firstOperand!! * secondOperand
            "/" -> if (secondOperand != 0.0) firstOperand!! / secondOperand else Double.NaN
            else -> secondOperand
        }

        currentInput.clear()
        currentInput.append(if (result.isNaN()) "Error" else removeTrailingZeros(result))
        firstOperand = null
        currentOperator = null
        updateDisplay()
    }
}

private fun toggleSign() {
    if (currentInput.isNotEmpty()) {
        val value = currentInput.toString().toDouble() * -1
        currentInput.clear()
        currentInput.append(removeTrailingZeros(value))
        updateDisplay()
    }
}

private fun applyPercent() {
    if (currentInput.isNotEmpty()) {
        val value = currentInput.toString().toDouble() / 100
        currentInput.clear()
        currentInput.append(removeTrailingZeros(value))
        updateDisplay()
    }
}

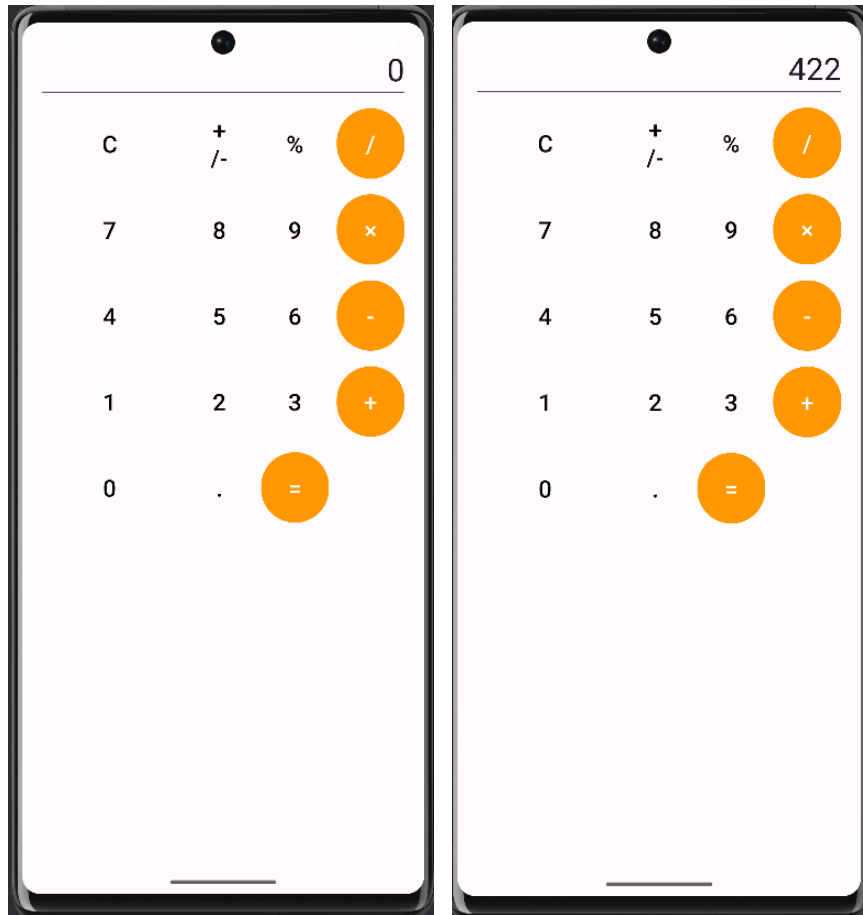
private fun updateDisplay() {
    val text = if (currentInput.isEmpty()) "0" else currentInput.toString()
    val selection = display.selectionEnd
    display.setText(text)
    if (selection <= text.length) {

```

```
        display.setSelection(selection)
    }
}

private fun removeTrailingZeros(number: Double): String {
    return if (number % 1 == 0.0) {
        String.format("%.0f", number)
    } else {
        number.toString()
    }
}
}
```

Output



Result:

The calculator components experiment has been successfully completed